

Sound waves propagate through many conducting structures in the ear before they are transduced into neuronal signals. When sound waves reach the fluid-filled cochlea, they are detected by hair cells lining the basilar membrane. The cochlear spiral can be modeled as a resonator system because each section of the basilar membrane is sensitive to a specific frequency (Figure 1).

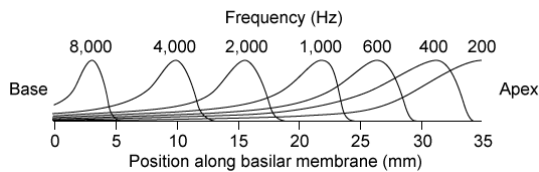


Figure 1 Amplitude pattern of the basilar membrane for different frequencies

Presbycusis (age-related hearing loss) results from a combination of factors that lessens an individual's mechanical and/or neurological sensitivity to sound. Pure tone audiometry (PTA) is used to determine an individual's ability to detect different sound frequencies and can be used to evaluate presbycusis. A "pure" tone is characterized by a single sinusoidal waveform. An audiogram plots the relative intensity required of a given frequency to be detected by the individual (Figure 2).

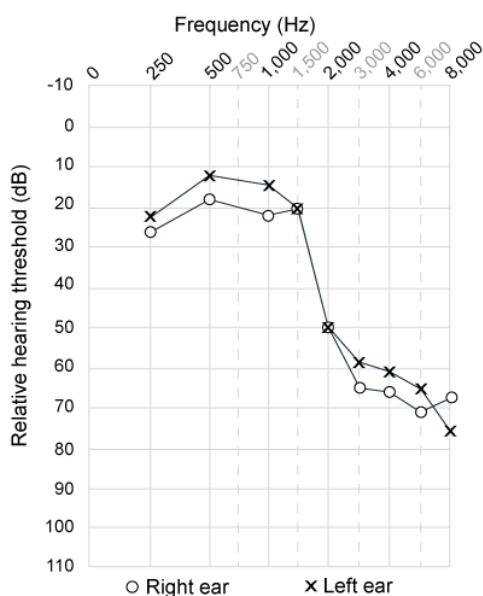


Figure 2 Audiogram of an elderly patient exhibiting presbycusis

In Figure 2, how does the hearing threshold intensity at 1,500 Hz compare to the threshold intensity at 2,000 Hz?

- ☐ A. It is 30 times less intense. [13%]
- ☐ B. It is 30 times more intense. [11%]
- ☒ C. It is 1,000 times less intense. [46%]
- ☐ D. It is 1,000 times more intense. [28%]

Correct

47% answered correctly

Explanation:

Decibel scale

Decibel (dB)	Relative intensity	Perceived loudness
-20	$\times 1/100$	$\times 1/4$
-10	$\times 1/10$	$\times 1/2$
0	$\times 1$	$\times 1$
+10	$\times 10$	$\times 2$
+20	$\times 100$	$\times 4$

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Intensity is defined as the amount of power (energy per unit time) delivered per unit area. The **intensity of sound** is related to its perceived loudness and is approximately logarithmic; a ten-fold increase in sound intensity is perceived to be twice as loud. The ratio of the intensities of two sounds is measured on a **base-10 logarithmic scale** with units of **decibels (dB)**:

$$\left[\text{dB} \right] = 10 \log \left(\frac{I_1}{I_2} \right)$$

For every increase in 10 dB, the ratio of the intensities increases by a factor of 10 (eg, +20 dB = $10^2 = 100$ times more intense). Conversely, the ratio of the intensities **decreases by a factor of 10** for each **decrease in 10 dB**.

From **Figure 2**, the hearing threshold at 1,500 Hz is 20 dB and the hearing threshold at 2000 Hz is 50 dB. Therefore, the relative intensity between the two frequencies in dB is the difference:

$$20 \text{ dB} - 50 \text{ dB} = -30 \text{ dB}$$

A sound that is **30 dB lower** is $10^3 = \mathbf{1,000 \text{ times less intense (Choice C)}}$. Alternatively, this relationship can be **resolved mathematically** using the definition of the logarithm.

(Choices A and B) Because the intensity of sound is measured on a logarithmic scale (not a linear scale), a change in 30 dB is not a 30-fold change in intensity.

(Choice D) Although the hearing threshold intensity for 1,500 Hz is *plotted* above the hearing threshold intensity for 2,000 Hz on the audiogram figure, it is important to note that the y-axis is inverted. The hearing threshold intensity at 1,500 Hz is thus lower than that for 2,000 Hz.

Educational objective:

The intensity of sound is measured on a logarithmic scale with units of decibels (dB). For each decrease in 10 dB, the sound intensity decreases by a factor of 10. Therefore, a decrease in 30 dB is equal to a 1,000-fold decrease in intensity.

Time spent: 0 sec

Subject:
Physics

QID: 400956

System:
Light and Sound

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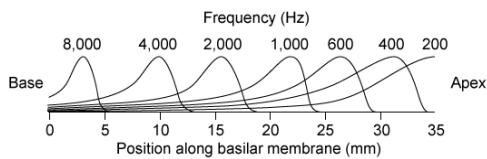


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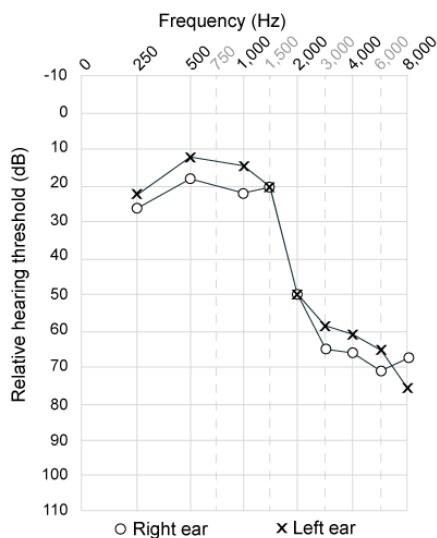
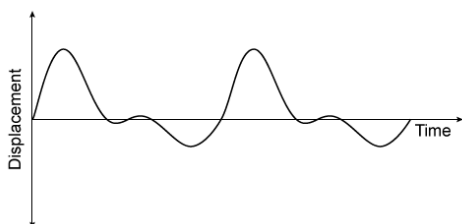


Figure 2 Audiogram of an elderly patient exhibiting presbycusis

Hearing loss due to loud environments can be mitigated with noise-canceling earphones, which generate a waveform that reduces the intensity of ambient noise through wave interference.



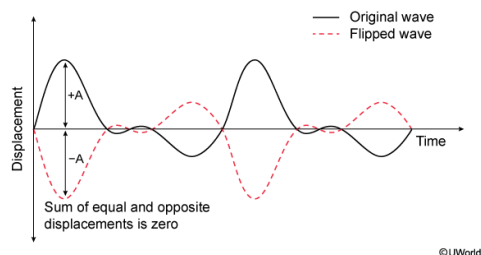
For the above sound wave, the waveform generated for complete noise cancellation will be:

- ☐ A. phase-shifted by half a wavelength. [22%]
- ☒ B. flipped across the x-axis. [71%]
- ☐ C. double the frequency. [2%]
- ☐ D. the exact same wave. [3%]

Correct

71% answered correctly

Explanation:



The intensity (perceived loudness) of a sound wave is related to its **amplitude (displacement)**. When more than one wave occupies the same space, **wave interference** occurs to **create one wave** that is the sum of the individual waves.

- Constructive interference occurs when the sum of the amplitudes results in a larger amplitude.
- **Destructive interference** occurs when the sum of the amplitudes results in a **smaller amplitude**.

Complete noise cancellation occurs when the generated sound wave destructively interferes with the original wave at all points, resulting in an amplitude of zero. A wave that is **flipped across the x-axis** will have **equal and opposite amplitudes** at each point along the wave. The sum of two equal and opposite amplitudes is zero, resulting in **total destructive interference** and canceling all noise.

(Choice A) The **phase of a wave** is the measure of the horizontal offset from a reference wave. When the given wave is superimposed with a copy of itself shifted by **half a wavelength**, it will be only partially canceled.

(Choice C) The frequency of a wave is the number of cycles that pass a fixed point per unit time. When a wave is superimposed with a copy of itself with **double the frequency**, both constructive and destructive interference can occur.

(Choice D) A wave summed with an exact copy of itself will result in constructive interference at all points throughout the wave.

Educational objective:

Constructive interference occurs when the sum of the amplitudes results in a larger amplitude.

Destructive interference occurs when the sum of the amplitudes results in a smaller amplitude. A wave

that is combined with a copy of itself that has been flipped across the x-axis would result in complete destructive interference.

Time spent: 0 sec

Subject:
Physics

QID: 400957

System:
Light and Sound

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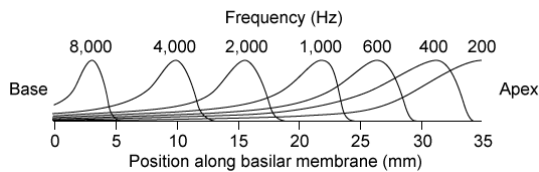


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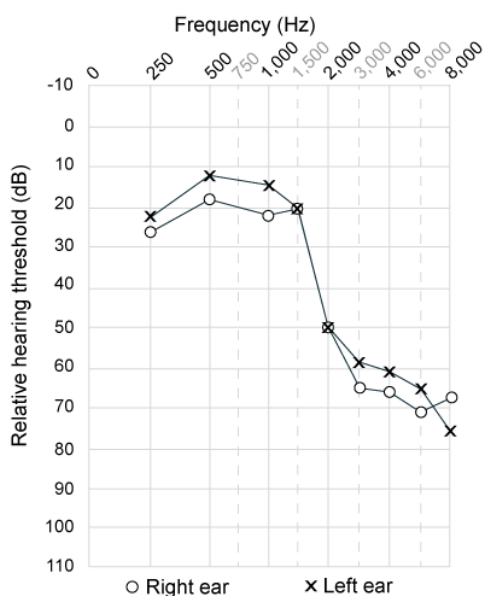


Figure 2 Audiogram of an elderly patient exhibiting presbycusis

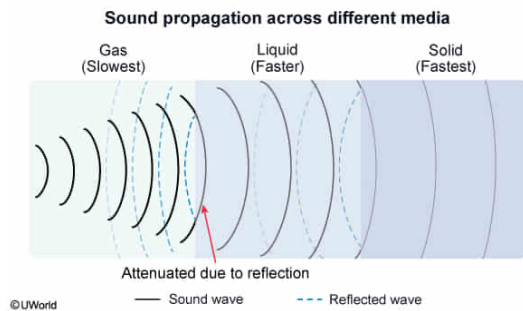
When sound passes from air to the tympanic membrane, what changes occur to the sound's intensity and velocity?

- ☐ A. Both increase [17%]
- ☐ B. Both decrease [23%]
- ☐ C. Intensity increases; velocity decreases [25%]
- ☒ D. Intensity decreases; velocity increases [33%]

Correct

34% answered correctly

Explanation:



A sound wave carries energy in the form of **longitudinal mechanical waves**, and the energy is related to the intensity of the sound wave. When sound crosses from **one medium to another**, a portion of the wave's **energy is reflected**. Therefore, sound waves lose energy (are attenuated) and their **intensity decreases** when passing from air to a solid structure, such as the **tympanic membrane**.

Wave velocity v is the product of wavelength λ and frequency f :

$$v = \lambda f$$

When a wave moves from one medium to another, its frequency does not change. Changes in wave velocity only cause the wavelength to change. The propagation **velocity of sound waves** depends on the properties of the medium:

- Velocity increases with temperature.
- Velocity is **slowest in gases**, faster in liquids, and **fastest in solids**.
- Within a phase of matter, velocity increases with stiffness and decreases with density.

Therefore, when sound passes from air to a solid (such as the tympanic membrane), its propagation **velocity increases**.

(Choices A and C) The intensity of sound waves decrease when passing from air to a solid structure because a portion of the wave is reflected at the interface between the two structures.

(Choice B) The velocity of sound increases as it moves from air to the tympanic membrane because sound velocity is fastest in solids.

Educational objective:

When sound moves from one medium to another, a portion of its energy is reflected, decreasing its intensity. The propagation velocity of sound depends on the medium and is slowest in gases, faster in liquids, and fastest in solids.

Time spent: 0 sec
Subject:
Physics

QID: 400958
System:
Light and Sound

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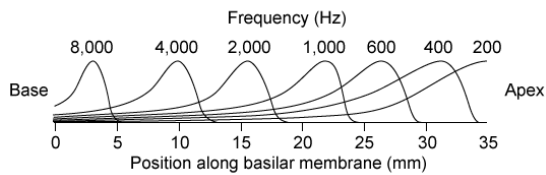


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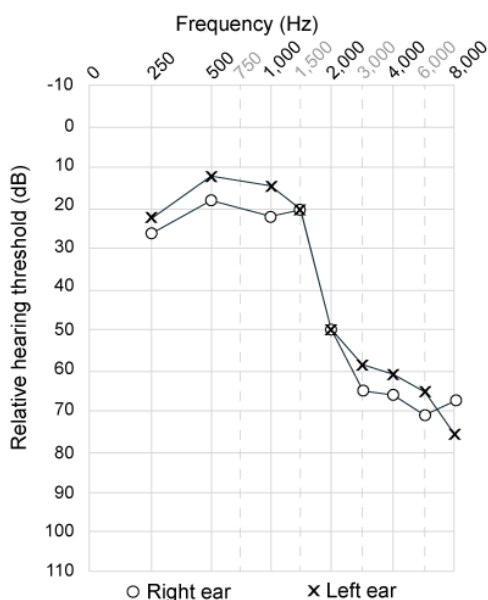


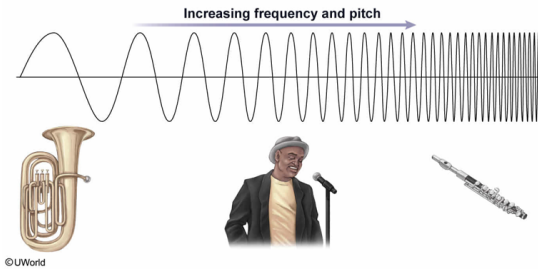
Figure 2 Audiogram of an elderly patient exhibiting presbycusis

A sound wave used in PTA that causes resonance close to the apex of the basilar membrane has a relatively:

- ☐ A. short wavelength. [16%]
- ☐ B. high intensity. [10%]
- ☒ C. low pitch. [67%]
- ☐ D. large phase shift. [5%]

Correct

67% answered correctly

Explanation:

According to the passage, the basilar membrane is modeled as a resonator system. The **resonant frequency** of an object is the frequency that causes **relatively large oscillations** compared to other frequencies. The resonant frequency depends on the physical characteristics and geometry of the object. As shown in Figure 1, high frequencies resonate near the base of the **basilar membrane**, and low frequencies resonate near the apex.

The frequency of a sound is associated with its perceived pitch (or tone). A sound with a high frequency is perceived to have a high pitch, and a sound with a **low frequency** (such as those near the **apex**) is perceived to have a **low pitch**.

(Choice A) From the wave velocity equation ($v = \lambda f$), wavelength is inversely proportional to frequency at a given velocity. Therefore, a low-frequency sound is characterized by a long wavelength.

(Choice B) The resonant frequency depends on the physical characteristics and geometry of the object. Resonance is independent from the intensity of the sound source.

(Choice D) A **phase shift** is the measure of the offset of a wave from a reference wave. A phase shift does not have an effect on resonance.

Educational objective:

The frequency of a sound is associated with its perceived pitch; high-frequency sounds have high pitches. The low-frequency sounds that cause resonance near the apex of the basilar membrane are characterized by low pitches.

Time spent: 0 sec
 Subject:
 Physics

QID: 400959
 System:
 Light and Sound

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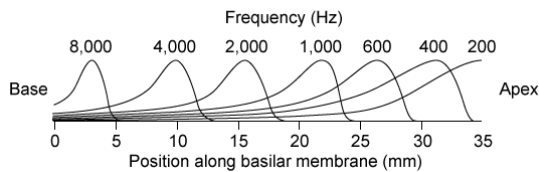


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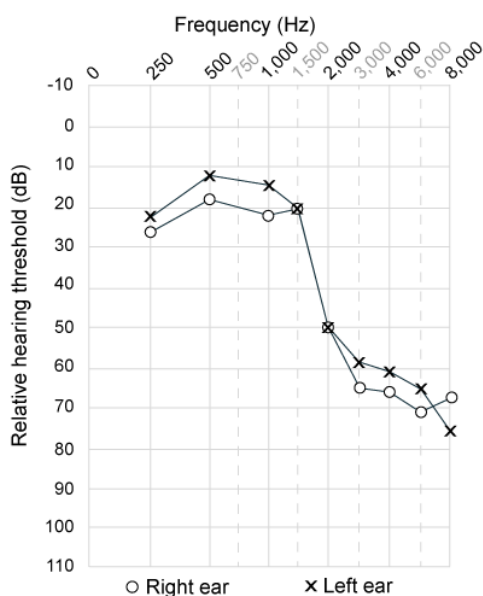


Figure 2 Audiogram of an elderly patient exhibiting presbycusis

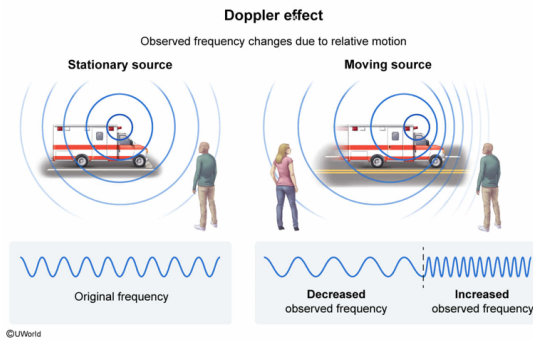
Audiometry can also be used to evaluate a subject's ability to localize a moving sound source. When the sound source is moving toward the subject, which of the following increases?

- ☐ A. The observed wave velocity [6%]
- ☐ B. The observed period [1%]
- ☒ C. The observed frequency [87%]
- ☐ D. The observed wavelength [3%]

Correct

88% answered correctly

Explanation:



When relative motion exists between a sound source and an observer, the observed wave differs from the original **frequency and wavelength**. This phenomenon is known as the **Doppler effect**.

When the sound **source moves toward the observer** (subject), the source generates each wave front at a location successively closer to the observer, shortening the observed wavelength. Because the propagation velocity is unchanged, the **observer detects** each wave at an **increased frequency** (rate) compared to the source frequency. The inverse relationship between frequency and wavelength can also be explained from the wave velocity equation ($v = \lambda f$).

(Choice A) The propagation velocity of sound waves depend only on the properties of the medium and is unaffected by the velocity of the source.

(Choice B) The period T of a wave measures how long one cycle (or one wavelength) takes to pass through a fixed point. Because period is the inverse of frequency ($T = 1/f$), the observed period will decrease if the sound source is moving toward the observer.

(Choice D) The observed wavelength would increase if the source and the observer are moving away from each other.

Educational objective:

When relative motion exists between a sound source and an observer, the observed frequency and wavelength differ from that of the original frequency and wavelength. This phenomenon is known as the Doppler effect. When the source moves toward the observer, the observed wavelength decreases and the observed frequency increases.

Time spent:0 sec

Subject:
Physics

QID:400960

System:
Light and Sound

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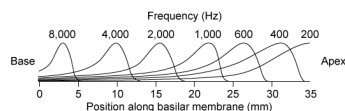


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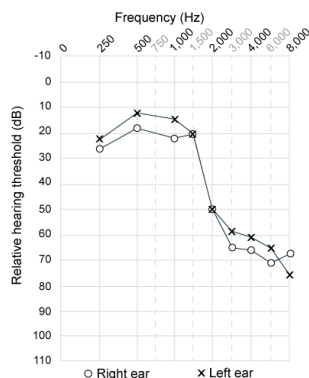


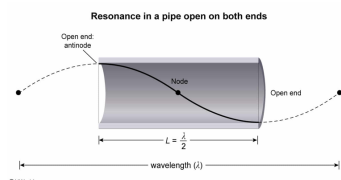
Figure 2 Audiogram of an elderly patient exhibiting presbycusis

The earliest stethoscope was simply a wooden pipe open at both ends. The fundamental frequency of a 0.25-m stethoscope will cause resonance at approximately what distance along the basilar membrane from its base? (Note: Use 350 m/s for the speed of sound in air.)

- ☐ A. 7 mm [13%]
- ☐ B. 16 mm [31%]
- ☒ C. 25 mm [46%]
- ☐ D. 32 mm [8%]

Correct
45% answered correctly

Explanation:



The **resonant frequency** of an object is the frequency that causes **relatively large oscillations** compared to other frequencies. The fundamental frequency is the lowest resonant frequency, and each subsequent **harmonic frequency** is a multiple of the fundamental frequency.

Resonant frequency waves are known as **standing waves** because the locations of the nodes and antinodes do not change. Nodes occur at the locations of zero displacement, and antinodes occur at the locations of maximum displacement.

The resonant frequency of a pipe depends on the length of the pipe and if the ends are closed. A closed end has a node (zero displacement) and an open end has an antinode (maximum displacement). The stethoscope is described as a **pipe open at both ends** for which the pipe's length is one half the wavelength of the fundamental frequency:

$$L = \frac{\lambda}{2}$$

Therefore, the resonant wavelength is twice the given length $L = 0.25$ m of the stethoscope:

$$\lambda = 2L = 0.5 \text{ m}$$

From the **wave velocity equation**, the velocity v of a wave is the product of its wavelength λ and frequency f .

$$v = \lambda f$$

Therefore, the frequency of the wave is the given speed of sound $v = 350$ m/s divided by $\lambda = 0.5$ m:

$$f = \frac{v}{\lambda} = \frac{350 \text{ m/s}}{0.5 \text{ m}} = 700 \text{ Hz}$$

From [Figure 1](#), a frequency of **700 Hz** would cause the basilar membrane to resonate around **25 mm** from its base.

(Choice D) For a [pipe closed on one end](#), the pipe's length is one-fourth the wavelength of the fundamental frequency: $L = \lambda/4$. This wavelength corresponds to a frequency of 350 Hz (around 32 mm).

Educational objective:

For a pipe open at both ends, the length is half the fundamental frequency wavelength ($L = \lambda/2$). From the wave velocity equation ($v = \lambda f$), frequency is calculated by dividing the wave's velocity by its wavelength.

Time spent: 0 sec

Subject:
Physics

QID: 400961

System:
Light and Sound